

Transportation Problem

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1. Introduction

The *Transportation Problem (TP)* is a special type of Linear Programming Problem (LPP) that deals with the transportation of goods from multiple sources (or origins) to multiple destinations in such a way that the total transportation cost is minimized (or profit is maximized).

It arises frequently in logistics, supply chain management, and production planning.

2. Mathematical Formulation

Suppose there are:

- m sources (origins): $S_1, S_2, \dots, S_m$, each having a supply a_i units.
- n destinations: $D_1, D_2, \dots, D_n$, each requiring b_j units.
- The cost of transporting one unit from O_i to D_j is c_{ij} .

	D_1	D_2	$\dots$	D_n	Supply $\downarrow$
S_1	c_{11}	c_{12}	$\dots$	c_{1n}	a_1
S_2	c_{21}	c_{22}	$\dots$	c_{2n}	a_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
S_m	c_{m1}	c_{m2}	$\dots$	c_{mn}	a_m
Demand $\rightarrow$	b_1	b_2	$\dots$	b_n	$\sum a_i \geq \sum b_j$

We define the decision variable:

x_{ij} = number of units transported from origin S_i to destination D_j .

Objective Function

The total transportation cost is:

$$\text{Minimize } Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}.$$

Subject to Constraints

$$\sum_{j=1}^n x_{ij} = a_i, \quad i = 1, 2, \dots, m \quad (\text{Supply constraints})$$

$$\sum_{i=1}^m x_{ij} = b_j, \quad j = 1, 2, \dots, n \quad (\text{Demand constraints})$$

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j, \quad (\text{Total supply should be greater than total demand})$$

$$x_{ij} \geq 0, \quad \forall i, j \quad (\text{Non-negativity})$$

3. Balanced and Unbalanced Problems

- The problem is called **balanced** if the total supply equals total demand:

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j.$$

- If the total supply is not equal to total demand, the problem is **unbalanced**. It can be converted into a balanced form by adding a *dummy source* or *dummy destination* with zero transportation cost.

Remark 1 *The necessary and sufficient condition to have the basic feasible solution of a transportation problem is: total supply equal to the total demand, i.e.,*

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j.$$

Remark 2 *Since the feasible region is bounded so in a TP an optimal solution always exist.*

Remark 3 *The number of basic variables in a TP is $m + n - 1$.*

4. Initial Basic Feasible Solution (IBFS)

To solve the transportation problem, we first find an *initial basic feasible solution*. Common methods:

1. North-West Corner Rule (NWCR)
2. Least Cost Method (LCM)
3. Vogel's Approximation Method (VAM)

Among these, Vogel's Approximation Method generally gives the best starting solution (closest to the optimum).

5. Optimality Test

After obtaining an IBFS, we test its optimality using:

- The **u-v method** (Modified Distribution Method, or MODI method).
- Compute the opportunity cost for each unoccupied cell:

$$\Delta_{ij} = c_{ij} - (u_i + v_j)$$

If all $\Delta_{ij} \geq 0$, the current solution is optimal.

6. Steps for Solving a Transportation Problem

1. Formulate the cost matrix $[c_{ij}]$ and supply/demand values.
2. Balance the problem if necessary.
3. Find an initial basic feasible solution using NWCR, LCM, or VAM.
4. Apply the $u-v$ (MODI) method to test for optimality.
5. If the solution is not optimal, adjust the allocations along the closed loop and repeat.
6. Stop when all opportunity costs $\Delta_{ij} \geq 0$.

Problem 1 A raw material is to be transported from two source point to three different destinations. The demand and supply quantity is given in the following table:

	Destination 1	Destination 2	Destination 3	Supply ↓
Source 1	Rs. 4/unit	Rs. 6/unit	Rs. 9/unit	100 units
Source 2	Rs. 7/unit	Rs. 5/unit	Rs. 3/unit	150 units
Demand →	80 units	70 units	100 units	

What amount should be transported to each destination to Minimize the transportation cost?

Solution: Let x_{ij} amount of raw material need to be transported from source i to destination j .

The objective function: $\min Z = 4x_{11} + 6x_{12} + 9x_{13} + 7x_{21} + 5x_{22} + 3x_{23}$

Subject to the constraint:

$$x_{11} + x_{12} + x_{13} = 100$$

$$x_{21} + x_{22} + x_{23} = 150$$

$$x_{11} + x_{21} = 80$$

$$x_{12} + x_{22} = 70$$

$$x_{13} + x_{23} = 100$$

7. North-West Corner Method (NWCN)

Objective: To find an initial basic feasible solution (IBFS) for the transportation problem.

1. Start at the North-West (top-left) corner cell of the transportation table, i.e., the cell (1,1).
2. Allocate to this cell the minimum of the available supply and demand:

$$x_{11} = \min(a_1, b_1)$$

3. Adjust the supply and demand:
 - Subtract the allocated quantity from both the corresponding supply and demand.
 - If supply becomes zero, move **down** to the next row.
 - If demand becomes zero, move **right** to the next column.
 - If both become zero, make an **arbitrary move**.
4. Repeat the procedure until all supplies and demands are satisfied.
5. The allocated cells represent the **basic variables**; their number should be $m + n - 1$.

Result: The set of allocations obtained gives an Initial Basic Feasible Solution (IBFS).

Problem 2 Find the initial basic feasible solution by North-West Corner Method.

	Destination 1	Destination 2	Destination 3	Supply ↓
Source 1	Rs. 10/unit	Rs. 35/unit	Rs. 15/unit	200 units
Source 2	Rs. 25/unit	Rs. 20/unit	Rs. 12/unit	150 units
Source 3	Rs. 15/unit	Rs. 19/unit	Rs. 21/unit	100 units
Demand →	100 units	200 units	150 units	

Solution:

Step I:

	Destination 1	Destination 2	Destination 3	Supply ↓
Source 1	10 100	35	15	200 100
Source 2	25	20	12	150
Source 3	15	19	21	100
Demand →	100	200	150	

Step II:

	Destination 1	Destination 2	Destination 3	Supply ↓
Source 1	10 100	35 100	15	200 100
Source 2	25	20	12	150
Source 3	15	19	21	100
Demand →	100	200 100	150	

Step III:

	Destination 1	Destination 2	Destination 3	Supply ↓
Source 1	10 100	35 100	15	200 100
Source 2	25	20 100	12	150 50
Source 3	15	19	21	100
Demand →	100	200 100	150	

Step IV:

	Destination 1	Destination 2	Destination 3	Supply ↓
Source 1	10 100	35 100	15	200 100
Source 2	25	20 100	12 50	150 50
Source 3	15	19	21	100
Demand →	100	200 100	150 100	

Step IV:

	Destination 1	Destination 2	Destination 3	Supply ↓
Source 1	10 100	35 100	15	200 100
Source 2	25	20 100	12 50	150 50
Source 3	15	19	21 100	100
Demand →	100	200 100	150 100	

Therefore, the initial BFS:

$$x_{11} = 100, x_{12} = 100, x_{22} = 100, x_{23} = 50, x_{33} = 100, \text{ Cost} = 9200.$$

Problem 3 Find the initial basic feasible solution by North-West Corner Method.

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	Rs. 10/unit	Rs. 2/unit	Rs. 20/unit	Rs. 11/unit	15 units
Source 2	Rs. 12/unit	Rs. 7/unit	Rs. 9/unit	Rs. 20/unit	25 units
Source 3	Rs. 4/unit	Rs. 14/unit	Rs. 16/unit	Rs. 18/unit	10 units
Demand →	5 units	15 units	15 units	15unit	

Answer: The initial BFS:

$$x_{11} = 5, x_{12} = 10, x_{22} = 5, x_{23} = 15, x_{24} = 5, x_{34} = 10, \text{ Cost} = 520.$$

8. Least Cost Method (LCM)

Objective: To find an initial basic feasible solution (IBFS) for the transportation problem by allocating to the least-cost cells first.

1. Identify the cell with the least transportation cost c_{ij} in the cost matrix.

2. Allocate to that cell the minimum of the available supply and demand:

$$x_{ij} = \min(a_i, b_j)$$

3. Adjust the supply and demand:

- Subtract the allocated quantity from both the corresponding supply and demand.
- If supply is exhausted, delete the corresponding row.
- If demand is satisfied, delete the corresponding column.
- If both become zero simultaneously, delete one (row or column) and set the other's supply or demand to zero.

4. Repeat the above steps with the reduced cost matrix until all supplies and demands are satisfied.

5. The number of basic cells (allocations) should be $m + n - 1$.

Result: The set of allocations obtained gives an Initial Basic Feasible Solution (IBFS), which generally provides a better total cost than the North-West Corner Method.

Problem 4 Find the initial basic feasible solution by Least Cost Method.

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	Rs. 10/unit	Rs. 2/unit	Rs. 20/unit	Rs. 11/unit	15 units
Source 2	Rs. 12/unit	Rs. 7/unit	Rs. 9/unit	Rs. 20/unit	25 units
Source 3	Rs. 4/unit	Rs. 14/unit	Rs. 16/unit	Rs. 18/unit	10 units
Demand →	5 units	15 units	15 units	15unit	

Answer: Step I:

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	10	2 15	20	11	15
Source 2	12	7	9	20	25
Source 3	4	14	16	18	10
Demand →	5	15	15	15	

Step II:

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	10	2 15	20	11	15
Source 2	12	7	9	20	25
Source 3	4 5	14	16	18	10 5
Demand →	5	15	15	15	

Step III:

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	10	2 15	20	11	15
Source 2	12	7	9 15	20	25 10
Source 3	4 5	14	16	18	10 5
Demand →	5	15	15	15	

Step IV:

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	10	2 15	20	11 0	15
Source 2	12	7	9 15	20	25 10
Source 3	4 5	14	16	18	10 5
Demand →	5	15	15	15	

Step V:

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	10	2 15	20	11 0	15
Source 2	12	7	9 15	20	25 10
Source 3	4 5	14	16	18 5	10 5
Demand →	5	15	15	15 10	

Step VI:

	Destination 1	Destination 2	Destination 3	Destination 4	Supply ↓
Source 1	10	2 15	20	11 0	15
Source 2	12	7	9 15	20 10	25 10
Source 3	4 5	14	16	18 5	10 5
Demand →	5	15	15	15 10	

The initial BFS:

$$x_{12} = 15, x_{14} = 0, x_{23} = 15, x_{24} = 10, x_{31} = 5, x_{34} = 5, \text{ Cost} = 475.$$

Since the value of a basic variable x_{14} is zero, which indicates the TP has degenerate basic solution.

Problem 5 Using Least Cost Method find the basic feasible solution of the following TP:

	D_1	D_2	D_3	D_4	Supply ↓
S_1	10	2	20	11	15
S_2	12	7	9	20	25
S_3	4	14	16	18	4
Demand →	5	15	15	15	

Answer: Since, the given TP is unbalance (supply is shorter), we add a dummy source point S_4 with supply 6 unit and transportation cost to each destination is 0. Hence, the given problem reduced to:

	D ₁	D ₂	D ₃	D ₄	Supply ↓
S ₁	10	2	20	11	15
S ₂	12	7	9	20	25
S ₃	4	14	16	18	4
S ₄	0	0	0	0	6
Demand →	5	15	15	15	

Now we will solve this by Least cost method:

	D ₁	D ₂	D ₃	D ₄	Supply ↓
S ₁	10	2 14	20	11 1	15 1
S ₂	12	7	9 15	20 10	25 10
S ₃	4	14	16	18 4	4
S ₄	0 5	0 1	0	0	6 1
Demand →	5	15 14	15	15 14 4	

The initial BFS:

$$x_{12} = 14, x_{14} = 1, x_{23} = 15, x_{24} = 10, x_{31} = 5, x_{34} = 4, \text{ Cost} = 446.$$

Problem 6 Using Least Cost Method find the basic feasible solution of the following TP:

	D ₁	D ₂	D ₃	D ₄	Supply ↓
S ₁	10	2	20	11	15
S ₂	12	7	9	20	25
S ₃	4	14	16	18	15
Demand →	5	15	15	15	

Answer: Since, the given TP is unbalance (demand is shorter), we add a dummy destination point D_5 with demand 5 unit and transportation cost to each destination is 0. Hence, the given problem reduced to:

	D ₁	D ₂	D ₃	D ₄	D ₅	Supply ↓
S ₁	10	2	20	11	0	15
S ₂	12	7	9	20	0	25
S ₃	4	14	16	18	0	15
Demand →	5	15	15	15	5	

Now we solve the above TP by LCM

	D ₁	D ₂	D ₃	D ₄	D ₅	Supply ↓
S ₁	10	2 10	20	11	0 5	15 10
S ₂	12	7 5	9 15	20 5	0	25 20 5
S ₃	4 5	14	16	18 10	0	15 10
Demand →	5	15 5	15	15 5	5	

The initial BFS:

$$x_{12} = 10, x_{22} = 5, x_{23} = 15, x_{24} = 5, x_{31} = 5, x_{34} = 10, \text{ Cost} = 490.$$

Second way: We assign the first cost at a_{35} cell:

	D ₁	D ₂	D ₃	D ₄	D ₅	Supply ↓
S ₁	10	2 15	20	11 0	0	15
S ₂	12	7	9 15	20 10	0	2510
S ₃	4 5	14	16	18 5	0 5	15 10 5
Demand →	5	15	15	15 10	5	

The initial BFS:

$$x_{12} = 10, x_{14} = 0, x_{23} = 15, x_{24} = 10, x_{31} = 5, x_{34} = 5, x_{35} = 5, \text{ Cost} = 475.$$

Here total number of basic variables = $7 = m + n - 1$, and all are not positive, so, this initial BFS is a degenerate solution.

8.1 Maximization type Transportation Problem

In some cases, the transportation cost between the source and destination will be replaced by the profit of selling product. In that case the transportation problem will be a maximization problem.

Convert the maximization problem into its equivalent minimization problem by multiplying the profit in each cell by (-1) . Now proceed, as in normal transportation problem. At the end multiply the final cost by (-1) to get the profit.

Problem 7 The profit of selling product from the source S_i to destination D_j is given into the following table:

	D ₁	D ₂	D ₃	Supply ↓
S ₁	10	35	15	200
S ₂	25	20	12	150
S ₃	15	19	21	100
Demand →	50	250	150	

Use the LCM to find out the initial basic feasible solution and corresponding maximum profit of the above TP.

Answer: The corresponding TP of minimization type is given by:

	D ₁	D ₂	D ₃	Supply ↓
S ₁	-10	-35	-15	200
S ₂	-25	-20	-12	150
S ₃	-15	-19	-21	100
Demand →	50	250	150	

Now proceed similar way as the previous problem, we can get the initial BFS

$$x_{12} = 200, x_{21} = 50, x_{22} = 50, x_{23} = 50, x_{33} = 100,$$

$$\text{Initial cost: } z = -11950$$

$$\text{Initial profit: } z = -(-11950) = 11950.$$

9. MODI (Modified Distribution) Method for Optimal Solution

Objective: To test and improve a basic feasible solution (BFS) of a transportation problem for optimality.

1. **Obtain an initial BFS:** Use North-West Corner, Least Cost, or Vogel's Approximation Method ensuring $m + n - 1$ basic cells.
2. **Compute potentials:** For each occupied cell (i, j) , write

$$u_i + v_j = c_{ij}$$

Assume one potential (say, $u_1 = 0$) and find all u_i, v_j .

3. **Compute opportunity costs:** For each unoccupied cell,

$$\Delta_{ij} = c_{ij} - (u_i + v_j)$$

4. **Test for optimality:** Since the transportation problem is inherently minimization type so:
 - If all $\Delta_{ij} \leq 0$, the solution is **optimal**.
 - If any $\Delta_{ij} > 0$, the solution is **not optimal**.
5. **Select the entering cell:** Choose the unoccupied cell with the most negative Δ_{ij} .
6. **Form a closed loop:** Start from the entering cell and move horizontally/vertically through occupied cells only, forming a loop. Mark alternate cells with '+' and '-', starting with a '+' in the entering cell.
7. **Find the leaving cell:** Among the cells marked '-', identify the one with the smallest allocation θ . This variable leaves the basis.
8. **Update allocations:** Add θ to the '+' cells and subtract θ from the '-' cells to get a new BFS.
9. **Repeat:** Go back to Step 2 until all $\Delta_{ij} \leq 0$. The final solution is optimal.

Result: The final transportation plan gives the minimum total transportation cost.

Problem 8 Find the optimal solution by MODI method from the following basic feasible solution:

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	10	2 14	20	11 2	16
S_2	12	7	9 15	20 10	25
S_3	4	14	16	18 4	4
S_4	2 5	3 1	4	1	6
Demand $\rightarrow$	5	15	15	16	

		$v_1=1$	$v_2=2$	$v_3=0$	$v_4=11$	
		D_1	D_2	D_3	D_4	
$u_1=0$	S_1	10 (-9)	2 + θ 14	20 (-20)	11 - θ 2	16
$u_2=9$	S_2	12 (-2)	7 (4)	9 15	20 10	25
$u_3=7$	S_3	4 (4)	14 (-5)	16 (-9)	18 4	4
$u_4=1$	S_4	2 5	3 - θ 1	4 (-3)	1 (11) + θ	6
		5	15	15	16	

Now the maximum θ that can be added and removed at the corners of the loop is such that the solution does not become infeasible. So we add $\min\{1, 2\} = 1$. Therefore, x_{42} cell leaves and x_{44} cell enters with allocation 1.

		$v_1=1$	$v_2=-9$	$v_3=-11$	$v_4=0$	
		D_1	D_2	D_3	D_4	
$u_1=11$	S_1	10 (2)	2 15	20 (-20)	11 1	16
$u_2=20$	S_2	12 (9)	7 (4)	9 15	20 10	25
$u_3=18$	S_3	4 (15) + θ	14 (-5)	16 (-9)	18 4 - θ	4
$u_4=1$	S_4	2 5 - θ	3 (-11)	4 (-14)	1 1 + θ	6
		5	15	15	16	

Now, $\min\{4, 5\} = 4$. Therefore, x_{34} cell leaves and x_{31} cell enters with allocation 4. Now,

		$v_1=12$	$v_2=2$	$v_3=0$	$v_4=11$	
		D_1	D_2	D_3	D_4	
$u_1=0$	S_1	10 (2)	2 15	20 (-20)	11 1	16
$u_2=9$	S_2	12 + θ (9)	7 (4)	9 15	20 10 - θ	25
$u_3=-8$	S_3	4 4	14 (-20)	16 (-24)	18 (-15)	4
$u_4=-10$	S_4	2 1 - θ	3 (-11)	4 (-14)	1 5 + θ	6
		5	15	15	16	

$\min\{10, 1\} = 1$. Therefore, x_{41} cell leaves and x_{21} cell enters with allocation 4.

		$v_1=3$	$v_2=2$	$v_3=0$	$v_4=11$	
		D_1	D_2	D_3	D_4	
$u_1=0$	S_1	10 (-7)	2 - θ 15 (-20)	20 11 + θ	16	
$u_2=9$	S_2	12 1 + θ (4)	7 15	9 9 - θ	25	
$u_3=1$	S_3	4 4 (-11)	14 15	16 (-15)	18 4	
$u_4=-10$	S_4	2 (-9)	3 (-11)	4 (-14)	1 6 16	
		5	15	15	16	

Now, $\min\{15, 9\} = 9$. Therefore, x_{24} cell leaves and x_{22} cell enters with allocation 9.

		$v_1=7$	$v_2=2$	$v_3=4$	$v_4=11$	
		D_1	D_2	D_3	D_4	
$u_1=0$	S_1	10 (-3)	2 6 (-16)	20 10 (-4)	11 10	16
$u_2=5$	S_2	12 1	7 9	9 15	20 (-4)	25
$u_3=-3$	S_3	4 4 (-15)	14 (-15)	16 (-15)	18 (-10)	4
$u_4=-10$	S_4	2 (-5)	3 (-11)	4 (-10)	1 6	6
		5	15	15	16	

The values of $(u_i + v_j - c_{ij})$ for all non basic variables is non-positive. So the optimality condition achieved. The optimal solution is given by

$$x_{12} = 6, x_{14} = 10, x_{21} = 1, x_{22} = 9, x_{23} = 15, x_{31} = 4, x_{44} = 6$$

$$\text{Optimal cost} = 354.$$

Problem 9 Find the initial basic feasible solution using Least Cost Method and find the Optima using MODI/u-v method. Does there exist an alternate optima?

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	16
S_2	4	7	9	1	37
S_3	4	4	7	5	34
Demand $\rightarrow$	16	18	31	25	

Answer:

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5 / $\swarrow$	3 / $\searrow$ 18	6 / $\swarrow$ 1	2 / $\swarrow$	19 +
S_2	4 / $\swarrow$	7 / $\swarrow$	9 / $\searrow$ 12	1 / $\searrow$ 25	27 12
S_3	4 / $\searrow$ 16	4 / $\swarrow$	7 / $\searrow$ 18	5 / $\swarrow$	34 18
Demand $\rightarrow$	16	18	31 30	25	90 90

The initial BFS of the given problem is

$$x_{12} = 18, x_{13} = 1, x_{23} = 12, x_{24} = 25, x_{31} = 16, x_{33} = 18$$

	$v_1=3$ D_1	$v_2=3$ D_2	$v_3=6$ D_3	$v_4=-2$ D_4	Supply $\downarrow$
$u_1=0$ S_1	5 (-2)	3 / $\searrow$ 18	6 / $\swarrow$ 1	2 (-4)	19
$u_2=3$ S_2	4 + θ	7	9 - θ	1	27
$u_3=1$ S_3	4 - θ	4 (-1)	7 + θ	5 (-6)	34
Demand $\rightarrow$	16	18	31	25	90 90

$\theta = \min\{12, 16\} = 12$. Therefore, x_{23} cell leaves and x_{21} cell enters with allocation 12. Hence, $u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the optimality condition achieved.

	$v_1=3$ D_1	$v_2=3$ D_2	$v_3=6$ D_3	$v_4=0$ D_4	Supply $\downarrow$
$u_1=0$ S_1	5 (-2)	3 / $\searrow$ 18	6 / $\swarrow$ 1	2 (-2)	19
$u_2=1$ S_2	4 / $\searrow$ 12	7 (-3)	9 (-2)	1 / $\searrow$ 25	27
$u_3=1$ S_3	4 / $\searrow$ 4	4 (0)	7 / $\searrow$ 30	5 (-4)	34
Demand $\rightarrow$	16	18	31	25	90 90

The optimal basic feasible solution is

$$x_{12} = 18, x_{13} = 1, x_{21} = 12, x_{24} = 25, x_{31} = 4, x_{33} = 30$$

Optimal cost = 359.

Alternating solution: In the last table, we can see the net evaluation ($u_i + v_j - C_{ij}$) is zero for a non basic variable x_{32} , which indicate the alternating solution exist. The alternate solution can be obtained by inserting the cell x_{32} .

	$v_1=3$	$v_2=3$	$v_3=6$	$v_4=0$	
	D_1	D_2	D_3	D_4	Supply $\downarrow$
$u_1=0$ S_1	5 (-2)	3 - θ 18	6 + θ 1	2 (-2)	19
$u_2=1$ S_2	4 12	7 (-3)	9 (-2)	1 25	37
$u_3=1$ S_3	4 4	4 + θ (0)	7 - θ 30	5 (-4)	34
Demand $\rightarrow$	16	18	31	25	90 90

$\theta = \min\{18, 30\} = 18$. Therefore, x_{12} cell leaves and x_{32} cell enters with allocation 18. Again, $u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the optimality condition achieved.

	$v_1=3$	$v_2=3$	$v_3=6$	$v_4=0$	
	D_1	D_2	D_3	D_4	Supply $\downarrow$
$u_1=0$ S_1	5 (-2)	3 (0)	6 19	2 (-2)	19
$u_2=1$ S_2	4 12	7 (-3)	9 (-2)	1 25	37
$u_3=1$ S_3	4 4	4 18	7 12	5 (-4)	34
Demand $\rightarrow$	16	18	31	25	90 90

The alternate optimal BFS is

$$x_{13} = 19, x_{21} = 12, x_{24} = 25, x_{31} = 4, x_{32} = 18, x_{33} = 12$$

Optimal cost = 359.

0.1 Degenerate alternating solutions

Problem 10 Find the initial basic feasible solution using Least Cost Method and find the Optima using MODI/u-v method. Does there exist an alternate optima? If yes find out the alternative solution.

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	4	6	8	8	40
S_2	6	8	6	7	60
S_3	5	7	6	8	50
Demand $\rightarrow$	20	30	50	50	

Answer: The initial basic feasible solution is given by

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	4 <u>20</u>	6 <u>20</u>	8	8	40
S_2	6	8 <u>10</u>	6 <u>50</u>	7	60
S_3	5	7	6	8 <u>50</u>	50
Demand $\rightarrow$	20	30	50	50	

Here total number of allocation is $5 \neq m + n - 1 = 6$, also the solution does not form a close loop. So, we introduce a new cell with allocation ϵ (very very small quantity) such that the new cell will not form a close loop with the existing allocation.

	$v_1=4$ D_1	$v_2=6$ D_2	$v_3=4$ D_3	$v_4=6$ D_4	Supply $\downarrow$
$u_1=0$ S_1	4 <u>20</u>	6 <u>20</u>	8 (-4)	8 (-2)	40
$u_2=2$ S_2	6 (0)	8 <u>-9</u> <u>10</u>	6 <u>+9</u> <u>50</u>	7 (1)	60
$u_3=2$ S_3	5 (1)	7 (1)	6 <u>+9</u> <u>50</u>	8 <u>-9</u> <u>50</u>	50
Demand $\rightarrow$	20	30	50	50	

$\theta = \min\{10, \epsilon\} = \epsilon$. Therefore, x_{33} cell leaves and x_{32} cell enters with allocation ϵ .

	$v_1=4$	$v_2=6$	$v_3=4$	$v_4=7$	
	D_1	D_2	D_3	D_4	Supply $\downarrow$
$u_1=0$ S_1	4 <u>20</u>	6 <u>20</u>	8 (-4)	8 (-1)	40
$u_2=2$ S_2	6 (0)	8 - θ <u>10</u>	6 <u>50</u>	7 + θ (2)	60
$u_3=1$ S_3	5 (0)	7 + θ <u>2</u>	6 (-1)	8 - θ <u>50</u>	50
Demand $\rightarrow$	20	30	50	50	

$\theta = \min\{10, 50\} = 10$. Therefore, x_{22} cell leaves and x_{24} cell enters with allocation 10.

	$v_1=4$	$v_2=6$	$v_3=6$	$v_4=7$	
	D_1	D_2	D_3	D_4	Supply $\downarrow$
$u_1=0$ S_1	4 <u>20</u>	6 <u>20</u>	8 (-2)	8 (-1)	40
$u_2=0$ S_2	6 (-2)	8	6 - θ <u>50</u>	7 + θ <u>10</u>	60
$u_3=1$ S_3	5 (0)	7 <u>10</u>	6 + θ <u>40</u>	8 - θ <u>40</u>	50
Demand $\rightarrow$	20	30	50	50	

$\theta = \min\{40, 50\} = 40$. Therefore, x_{34} cell leaves and x_{33} cell enters with allocation 40.

	$v_1=4$	$v_2=6$	$v_3=5$	$v_4=6$	
	D_1	D_2	D_3	D_4	Supply $\downarrow$
$u_1=0$ S_1	4 <u>20</u>	6 <u>20</u>	8 (-2)	8 (-2)	40
$u_2=1$ S_2	6 (-1)	8 (-1)	6 <u>10</u>	7 <u>50</u>	60
$u_3=1$ S_3	5 (0)	7 <u>10</u>	6 <u>40</u>	8 (-1)	50
Demand $\rightarrow$	20	30	50	50	

Hence, $u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the optimality condition achieved. The optimal basic feasible solution is

$$x_{11} = 20, x_{12} = 20, x_{23} = 10, x_{24} = 50, x_{32} = 10, x_{33} = 40$$

$$\text{Optimal cost} = 920.$$

Alternate Solution: we can see the net evaluation $(u_i + v_j - C_{ij})$ is zero for a non basic variable x_{31} , which indicate the alternating solution exist. The alternate solution can be obtained by inserting the cell x_{31} .

		$v_1=4$	$v_2=6$	$v_3=5$	$v_4=6$	
		D_1	D_2	D_3	D_4	Supply $\downarrow$
$u_1=0$	S_1	4	6	8	8	40
			$\frac{-\theta}{20}$	$\frac{+\theta}{20}$	(-2)	(-2)
$u_2=1$	S_2	6	8	6	7	60
		(-1)	(-1)	10	50	
$u_3=1$	S_3	5	7	6	8	50
		(0)	10	40	(-1)	
	Demand $\rightarrow$	20	30	50	50	

$\theta = \min\{10, 20\} = 10$. Therefore, x_{22} cell leaves and x_{24} cell enters with allocation 10.

		$v_1=4$	$v_2=6$	$v_3=5$	$v_4=6$	
		D_1	D_2	D_3	D_4	Supply $\downarrow$
$u_1=0$	S_1	4	6	8	8	40
		10	30	(-2)	(-2)	
$u_2=1$	S_2	6	8	6	7	60
		(-1)	(-1)	10	50	
$u_3=1$	S_3	5	7	6	8	50
		10	(0)	40	(-1)	
	Demand $\rightarrow$	20	30	50	50	

$u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the optimality condition achieved. The alternate optimal BFS is

$$x_{11} = 10, x_{12} = 30, x_{23} = 10, x_{24} = 50, x_{31} = 10, x_{33} = 40$$

Optimal cost = 920.

0.2 Restricted transportation problem

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	19
S_2	8	7	9	1	37
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	31	25	

Problem 11 Consider the above table and find the initial basic feasible solution using Least Cost Method and find the Optima using MODI/u-v method under the restriction that **the** transportation from S_2 to D_4 is not possible.

Answer: Since, the transportation from S_2 to D_4 is not possible, we assign the corre-

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	19
S_2	8	7	9	1	37
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	31	25	

sponding cost M (very large number).

	$v_1=1$ D_1	$v_2=2$ D_2	$v_3=4$ D_3	$v_4=2$ D_4	Supply $\downarrow$
$u_1=0$ S_1	5 (-4)	3 (-1)	6 (-2)	2	19
$u_2=5$ S_2	8 (-2)	7	9	M (7-2)	37
$u_3=3$ S_3	4	10 (-5)	7	5	34
Demand $\rightarrow$	16	18	31	25	

$u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the optimality condition achieved. The alternate optimal BFS is

$$x_{14} = 19, x_{22} = 18, x_{23} = 19, x_{31} = 16, x_{33} = 12, x_{34} = 6$$

$$\text{Optimal cost} = 2(19) + 7(18) + 9(19) + 4(16) + 7(12) + 5(6) = 513.$$

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	19
S_2	8	7	9	1	37
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	31	25	

Problem 12 Consider the above table and find the initial basic feasible solution using Least Cost Method and find the Optima using MODI/u-v method under the restriction that **the** transportation from S_2 to D_4 should be 20 units.

Answer: Due to the given restriction, we first assign 20 units in x_{24} cell.

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	19
S_2	8	7	9	1	17
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	31	5	

Now we solve the TP by LCM to find the initial basic feasible solution.

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5 /	3 $\boxed{14}$	6 /	2 $\boxed{5}$	19 14
S_2	8 /	7 $\boxed{4}$	9 $\boxed{13}$	1 $\boxed{20}$	17 13
S_3	4 $\boxed{16}$	10 /	7 $\boxed{18}$	5 /	34 18
Demand $\rightarrow$	16	18 4	31 13	5	

Now we apply MODI method to find the optimal solution.

	$v_1=2$ D_1	$v_2=3$ D_2	$v_3=5$ D_3	$v_4=2$ D_4	Supply $\downarrow$
$u_1=0$ S_1	5 (-3)	3 $\boxed{14}$	6 (-1)	2 $\boxed{5}$	19
$u_2=4$ S_2	8 (-2)	7 $\boxed{4}$	9 $\boxed{13}$	M $\boxed{6-M}$	17
$u_3=2$ S_3	4 $\boxed{16}$	10 (-5)	7 $\boxed{18}$	5 (-1)	34
Demand $\rightarrow$	16	18	31	5	

Since, $u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the current solution is an optimal solution, which is given by

$$x_{12} = 14, x_{14} = 5, x_{22} = 4, x_{23} = 13, x_{24} = 20, x_{31} = 16, x_{33} = 18$$

$$\text{Optimal cost} = 3(14) + 2(5) + 7(4) + 9(13) + 4(16) + 7(18) + 1(20) = 407.$$

Problem 13 Consider the following table and find the initial basic feasible solution using Least Cost Method and find the Optima using MODI/u-v method under the restriction that the transportation from S_2 to D_3 at least 20 units.

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	19
S_2	8	7	9	1	37
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	31	25	

Answer: Due to the given restriction, we first assign 20 units in x_{24} cell. Now we solve

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	19
S_2	8	7	9	1	17
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	11	35	

the TP by LCM to find the initial basic feasible solution.

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5 /	3 	6 /	2 8	19 11
S_2	8 —	7 —	9 (20)	1 17	37 17
S_3	4 16	10 7	7 	5 /	34 18 7
Demand $\rightarrow$	16	18 7	11	25 8	

Now we apply MODI method to find the optimal solution.

	$v_1 = -3$ D_1	$v_2 = 3$ D_2	$v_3 = 0$ D_3	$v_4 = 2$ D_4	Supply $\downarrow$
$u_1 = 0$ S_1	5 (-2)	3 +0	6 (-6)	2 -0	19
$u_2 = -1$ S_2	8 (-12)	7 (-5)	9 (-10)	1 (-2)	17
$u_3 = 7$ S_3	4 (16)	10 (-7)	7 (11)	5 (4)	34
Demand $\rightarrow$	16	18	11	25	

$\theta = \min\{7, 8\} = 7$. Therefore, x_{33} cell leaves and x_{34} cell enters with allocation 7.

	$v_1 = 1$ D_1	$v_2 = 3$ D_2	$v_3 = 4$ D_3	$v_4 = 2$ D_4	Supply $\downarrow$
$u_1 = 0$ S_1	5 (-4)	3 (11)	6 (-2)	2 (1)	19
$u_2 = -1$ S_2	8 (-8)	7 (-5)	9 (-6)	1 (17)	17
$u_3 = 3$ S_3	4 (16)	10 (-4)	7 (11)	5 (7)	34
Demand $\rightarrow$	16	18	11	25	

Since, $u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the current solution is an optimal solution, which is given by

$$x_{12} = 11, x_{14} = 1, x_{24} = 17, x_{31} = 16, x_{33} = 11, x_{34} = 7$$

$$\text{Optimal cost} = 3(11) + 2(1) + 1(17) + 4(16) + 7(11) + 5(7) + 9(20) = 429.$$

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	19
S_2	8	7	9	1	17
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	31	25	

Problem 14 Consider the above table and find the initial basic feasible solution using Least Cost Method and find the Optima using MODI/u-v method under the restriction that **the** transportation from S_1 to D_2 is atmost 15 units.

Answer: Since the transportation from S_1 to D_2 is atmost **15** units, we rewrite the table in the following way:

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5	3	6	2	15
S'_1	5	M	6	2	4
S_2	8	7	9	1	37
S_3	4	10	7	5	34
Demand $\rightarrow$	16	18	31	25	

Now we will find the initial BFS from the above table by LCM.

	D_1	D_2	D_3	D_4	Supply $\downarrow$
S_1	5 /	3 $\sqrt{15}$	6 /	2 /	15
S'_1	5 /	M	6 $\sqrt{4}$	2 /	4
S_2	8 /	7 $\sqrt{3}$	9 $\sqrt{9}$	1 $\sqrt{25}$	37 $\sqrt{12}$
S_3	4 $\sqrt{16}$	10 /	7 $\sqrt{18}$	5 /	34 $\sqrt{18}$
Demand $\rightarrow$	16	18	31	25	

We use the MODI method to check the optimality of the initial BFS.

	$v_1=2$ D_1	$v_2=3$ D_2	$v_3=5$ D_3	$v_4=-3$ D_4	Supply $\downarrow$
$u_1=0$ S_1	5 (-3)	3 $\sqrt{15}$	6 (-1)	2 (-5)	15
$u_2=1$ S'_1	5 (-2)	M $(4-1)$	6 $\sqrt{4}$	2 (-4)	4
$u_3=4$ S_2	8 (-2)	7 $\sqrt{3}$	9 $\sqrt{9}$	1 $\sqrt{25}$	37
$u_4=2$ S_3	4 $\sqrt{16}$	10 (-5)	7 $\sqrt{18}$	5 (-6)	34
Demand $\rightarrow$	16	18	31	25	

Since, $u_i + v_j - c_{ij} \leq 0$ for all non basic variables. So, the current solution is an optimal solution, which is given by

$$x_{12} = 15, x_{13} = 4, x_{22} = 3, x_{23} = 9, x_{24} = 25, x_{31} = 16, x_{33} = 18$$

$$\text{Optimal cost} = 3(15) + 6(4) + 7(3) + 9(9) + 1(25) + 4(16) + 7(18) = 386.$$